

AIG-Models × PAVS — Collaboration Landscape

Meet & Greet · Prateek Prabhanjan's team · **Hyperloom** — agentic kernel optimization on AMD HW

What AIG-Models brings

- **Hyperloom** — automated kernel optimization pipeline on AMD silicon
- **Kernel Generation** expertise (Kush, Ravi)
- **Synthetic Data Generation** — Genesis-style simulators for training
- **World Action Robotic Models** — being developed & optimized now; trending next few months

Deep expertise in model internals: layers, kernels, quantization — and the tooling to automate it.

What PAVS brings

- **Edge HW coverage** — GPK, krk2e, PNR (Strix, Krackan, Panther Lake)
- **PAI SDK** — models already packaged and ready for optimization
- **AMD-ROS pipelines** — robotics middleware stack on AMD HW
- **AMD-cUROBO** — motion planning (cuRobo ported to ROCm)
- **AARTS / Vertical Software** — research → production-grade SDK

The deployment end: Edge HW, robotics pipelines, and the SDK that ships models to customers.



together we advance₁

AIG optimizes models. PAVS deploys them. **The two teams are the two halves of the same pipeline.**

Four Collaboration Tracks

Where the teams overlap and what moves next

① Hyperloom → Edge HW

- Extend Hyperloom to **GPK · krk2e · PNR**
- Start from GEEK system; optimize **PAI SDK models** on NPU + GPU
- Amol · Vaibhav · Karthik leading

② World Action Robotic Models

- AIG developing & optimizing **World Action models** now
- Integrate and **benchmark on Edge HW** (GPU · NPU)
- Srini leading integration from PAVS side

③ Synthetic Data & Kernel Gen

- AIG running **Genesis-style simulators** for synthetic data
- PAVS robotics / sim-to-real work can **directly benefit**
- Kush · Ravi (AIG) — connect with PAVS sim team

④ Assets Exchange — Kick-off Session

- AIG wants to learn: **AMD-ROS** pipelines, **AMD-cUROBO** motion planner
- PAVS assets from **PAVS model zoo** for AIG model dev
- Action: schedule robotics kick-off session

Near-term action: Amol/Vaibhav/Karthik start Hyperloom on GPK · Srini aligns on World Action models · Schedule robotics assets session with AIG